

In the Clinic®

Cellulitis and Soft Tissue Infections

Cellulitis and soft tissue infections are a diverse group of diseases that range from uncomplicated cellulitis to necrotizing fasciitis. Management of predisposing conditions is the primary means of prevention. Cellulitis is a clinical diagnosis and thus is made on the basis of history and physical examination. Imaging may be helpful for characterizing purulent soft tissue infections and associated osteomyelitis. Treatment varies according to the type of infection. The foundations of treatment are drainage of purulence and antibiotics, the latter targeted at the infection's most likely cause.

Prevention

Diagnosis

Treatment

Practice Improvement

CME/MOC activity available at [Annals.org](https://www.annals.org).

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1. Miller LG, Eisenberg DF, Liu H, Chang CL, Wang Y, Luthra R, et al. Incidence of skin and soft tissue infections in ambulatory and inpatient settings, 2005-2010. *BMC Infect Dis.* 2015;15:362. [PMID: 26293161]
2. Björnsdóttir S, Gottfredsson M, Thórisdóttir AS, Gunnarsson GB, Ríkardsdóttir H, Kristjánsson M, et al. Risk factors for acute cellulitis of the lower limb: a prospective case-control study. *Clin Infect Dis.* 2005;41:1416-22. [PMID: 16231251]
3. Quirke M, Ayoub F, McCabe A, Boland F, Smith B, O'Sullivan R, et al. Risk factors for nonpurulent leg cellulitis: a systematic review and meta-analysis. *Br J Dermatol.* 2017;177:382-394. [PMID: 27864837]
4. Al-Niaimi F, Cox N. Cellulitis and lymphoedema: A vicious cycle. *J Lymphoedema.* 2009;4:38-42.
5. Singh N, Armstrong DG, Lipsky BA. Preventing foot ulcers in patients with diabetes. *JAMA.* 2005;293:217-28. [PMID: 15644549]
6. Dire DJ, Coppola M, Dwyer DA, Lorette JJ, Karr JL. Prospective evaluation of topical antibiotics for preventing infections in uncomplicated soft tissue wounds repaired in the ED. *Acad Emerg Med.* 1995;2:4-10. [PMID: 7606610]
7. Stevens DL, Bisno AL, Chambers HF, Dellinger EP, Goldstein EJ, Gorbach SL, et al; Infectious Diseases Society of America. Practice guidelines for the diagnosis and management of skin and soft tissue infections: 2014 update by the Infectious Diseases Society of America. *Clin Infect Dis.* 2014;59:e10-52. [PMID: 24973422]
8. Oh CC, Ko HC, Lee HY, Safdar N, Maki DG, Chlebicki MP. Antibiotic prophylaxis for preventing recurrent cellulitis: a systematic review and meta-analysis. *J Infect.* 2014;69:26-34. [PMID: 24576824]
9. Thomas KS, Crook AM, Nunn AJ, Foster KA, Mason JM, Chalmers JR, et al; U.K. Dermatology Clinical Trials Network's PATCH I Trial Team. Penicillin to prevent recurrent leg cellulitis. *N Engl J Med.* 2013;368:1695-703. [PMID: 23635049]

Bacterial skin and soft tissue infections (SSTIs) are a diverse group of diseases ranging from uncomplicated cellulitis, which can be treated in the outpatient setting with oral antibiotics, to necrotizing fasciitis, a life-threatening condition that requires emergent surgical debridement (**Table 1**). In the early years of the 21st century, the incidence of SSTIs in the United States increased dramatically, coincident with the emergence of community-associated methicillin-resistant *Staphylococcus aureus* (MRSA). More recently, the incidence of SSTIs has

stabilized but remains high, at 48.5 cases per 1000 person-years—2-fold higher than urinary tract infections and 10-fold higher than pneumonia (1). Challenges for clinicians include the lack of reliable microbiologic and laboratory diagnostics for cellulitis that makes it difficult to differentiate from noninfectious mimics. The high prevalence of drug-resistant organisms and the increasing number of immunocompromised patients in the age of transplantation and use of biologic immunosuppressives add to the complexity of treating these infections.

Prevention

What factors increase the risk for cellulitis and soft tissue infections?

Disruption of the skin barrier provides a critical portal for pathogens to enter the skin and soft tissues (see the **Box**: Risk Factors for Cellulitis and Soft Tissue Infections). Both traumatic injuries (such as penetrating wounds, injection drug use, abrasions, and bites) and non-traumatic skin disruptions (such as ulcers, dermatitis, and tinea pedis) predispose to SSTIs. Impaired venous and lymphatic drainage (prior saphenectomy, lymph node dissection, lymphedema, venous stasis, and obesity) as well as peripheral artery disease are important risk factors. Patients with comorbid conditions that predispose them to infection, particularly diabetes, are also at increased risk. A history of cellulitis greatly increases the risk for subsequent episodes (2, 3).

How can patients decrease their risk for cellulitis and soft tissue infections?

Prevention should focus on management of predisposing conditions. Careful attention

should be paid to wound care; treatment of tinea pedis; and preventing dry, cracked skin. Aggressive management of lymphedema may decrease the risk for recurrent cellulitis (4). Predisposing factors should be identified and treated at the time of initial diagnosis to decrease the risk for recurrence.

For diabetic patients, regular foot examinations, screening for diabetic neuropathy, debridement of calluses, and use of orthotic footwear can decrease risk for ulcers and subsequent infection (5). All patients should be provided general foot care recommendations and taught how to inspect their feet regularly.

Acute traumatic wounds should be copiously irrigated, foreign objects removed, and devitalized tissues debrided. For highly contaminated wounds, high-pressure irrigation should be considered (>7 pounds per square inch can be achieved using a 10- to 50-mL syringe and a splash guard). Contaminated wounds should be left open to heal by secondary in-

Table 1. Definitions of Skin and Soft Tissue Infections

Cellulitis	Acute infection of skin involving deep dermis and subcutaneous fat
Erysipelas	More superficial infection of the skin, involving the lymphatics; characterized by a tender, erythematous plaque with well-demarcated borders
Folliculitis	Superficial infection of the hair follicle with purulence in the epidermis
Furuncle	Infection of the hair follicle with associated small subcutaneous abscess
Carbuncle	A cluster of furuncles
Cutaneous abscess	Localized collection of pus within the dermis and deeper skin tissues
Pyomyositis	Purulent infection of skeletal muscle, often with abscess formation
Impetigo	Superficial infection of the skin characterized by pustules or vesicles that progress to crusting or bullae
Ecthyma	A deeper variant of impetigo; begins as vesicles/pustules and evolves into "punched-out"-appearing ulcers
Gas gangrene	Necrotizing infection involving muscle; also known as clostridial myonecrosis
Necrotizing fasciitis	Aggressive infection of the subcutaneous tissue that spreads along fascial planes

tention or by delayed primary closure. Topical antimicrobial agents (triple antibiotic ointment, neomycin, mupirocin) have been shown to decrease the rate of subsequent infection in uncomplicated wounds treated in the emergency department (6).

When should antimicrobial prophylaxis be considered for recurrent cellulitis?

Prophylactic antibiotics can prevent recurrence in patients with frequent episodes of cellulitis and may be considered for patients who have 3 to 4 episodes per year despite attempts to treat or control predisposing factors (7). Regimens that include oral penicillin, intramuscular penicillin, and erythromycin have been investigated.

A systematic review and meta-analysis of 5 trials found a pooled risk ratio of 0.46 with prophylaxis (8). The PATCH I randomized trial of 274 patients with at least 1 prior recurrence of cellulitis found a nearly 50% reduction over 12 months with a twice-daily oral dose of 250 mg of penicillin versus placebo. However, the protective effect diminished progressively once penicillin was stopped (9).

Which strategies decrease cellulitis and soft tissue infections after mammalian bites?

Initial management of bite wounds includes thorough cleaning with copious irrigation and povidone-iodine. Puncture wounds should be carefully explored to identify foreign objects and injury to underlying structures. Nonviable tissue should be debrided. Primary closure of bite wounds is generally not recommended, except for facial wounds, which may be closed if copiously irrigated and treated with prophylactic antibiotics.

Primary closure of uncomplicated dog bite wounds on the body or face can also be considered based on a 2014 randomized trial in which the infection rate was similar to that of nonclosure, and the cosmetic appearance in the primary suturing group was significantly improved (10).

Guidelines from the Infectious Diseases Society of America (IDSA) recommend administration of early prophylactic antibiotics to patients with high-risk factors (immunocompromise,

Risk Factors for Cellulitis and Soft Tissue Infections

Disruption of the skin barrier

- Traumatic: Lacerations, recent surgery, burns, abrasions, crush injuries, open fractures, injection drug use, human and animal bites, insect bites
- Nontraumatic: Ulcers, tinea pedis, dermatitis, toe web intertrigo
- Impaired drainage
- Axillary and pelvic lymph node dissection
- Saphenous vein harvesting
- Lymphedema
- Obesity
- Chronic venous insufficiency

Peripheral artery disease

Conditions that predispose to infection

- Diabetes
- Cirrhosis
- Neutropenia
- HIV
- Transplantation and immunosuppressive medications

Homelessness

History of cellulitis

10. Paschos NK, Makris EA, Gantsos A, Georgoulis AD. Primary closure versus non-closure of dog bite wounds: a randomised controlled trial. *Injury*. 2014;45:237-40. [PMID: 23916901]
11. Jaendl M, Oberleitner G, Endler G, Thallinger C, Kovar FM. Management of bite wounds in children and adults: an analysis of over 5000 cases at a level I trauma centre. *Wien Klin Wochenschr*. 2016;128:367-75. [PMID: 26659907]
12. Anderson DJ, Podgorny K, Berrios-Torres SI, Bratzler DW, Dellinger EP, Greene L, et al. Strategies to prevent surgical site infections in acute care hospitals: 2014 update. *Infect Control Hosp Epidemiol*. 2014;35 Suppl 2:S66-88. [PMID: 25376070]

13. Schweizer M, Perencevich E, McDanel J, Carson J, Formanek M, Hafner J, et al. Effectiveness of a bundled intervention of decolonization and prophylaxis to decrease Gram positive surgical site infections after cardiac or orthopedic surgery: systematic review and meta-analysis. *BMJ*. 2013;346:f2743. [PMID: 23766464]
14. Simor AE, Phillips E, McGeer A, Konvalinka A, Loeb M, Devlin HR, et al. Randomized controlled trial of chlorhexidine gluconate for washing, intranasal mupirocin, and rifampin and doxycycline versus no treatment for the eradication of methicillin-resistant *Staphylococcus aureus* colonization. *Clin Infect Dis*. 2007;44:178-85. [PMID: 17173213]
15. Ellis MW, Schlett CD, Millar EV, Wilkins KJ, Crawford KB, Morrison-Rodriguez SM, et al. Hygiene strategies to prevent methicillin-resistant *Staphylococcus aureus* skin and soft tissue infections: a cluster-randomized controlled trial among high-risk military trainees. *Clin Infect Dis*. 2014;58:1540-8. [PMID: 24633684]
16. Robicsek A, Beaumont JL, Thomson RB Jr, Govindarajan G, Peterson LR. Topical therapy for methicillin-resistant *Staphylococcus aureus* colonization: impact on infection risk. *Infect Control Hosp Epidemiol*. 2009;30:623-32. [PMID: 19496730]
17. Fritz SA, Hogan PG, Hayek G, Eisenstein KA, Rodriguez M, Epplin EK, et al. Household versus individual approaches to eradication of community-associated *Staphylococcus aureus* in children: a randomized trial. *Clin Infect Dis*. 2012;54:743-51. [PMID: 22198793]
18. Weng QY, Raff AB, Cohen JM, Gunasekera N, Okhovat JP, Vedak P, et al. Costs and Consequences Associated With Misdiagnosed Lower Extremity Cellulitis. *JAMA Dermatol*. 2016. [PMID: 27806170]
19. Goldstein E, Anaya DA, Dellinger EP. Necrotizing soft tissue infection: diagnosis and management. *Clin Infect Dis*. 2007;44:705-10.

asplenia, advanced liver disease, preexisting or resultant edema) or high-risk bite wounds (moderate to severe injury especially to the hands or face, or injuries that may have penetrated the periosteum or joint capsule) (7). Human and cat bites are notoriously infectious, and cat bites are particularly associated with osteomyelitis because of the deep penetration of cat teeth (11).

Prophylactic antibiotics should be given for all closed-fist injuries, unless the skin has not been penetrated, and for puncture wounds caused by cat bites. The antibiotic should have both aerobic and anaerobic activity and include *Pasteurella* coverage for animal bites and *Eikenella* for human bites; suggested regimens include amoxicillin-clavulanate. If the patient is allergic to penicillin, clindamycin plus a levofloxacin or moxifloxacin, which has anaerobic coverage, can be used. *Capnocytophaga canimorsus* is a pathogen that can rarely cause overwhelming infection after animal bites (most frequently from dogs), particularly in patients with asplenia or cirrhosis, and is sensitive to both amoxicillin-clavulanate and clindamycin.

In addition to antibiotic prophylaxis, tetanus vaccination should be considered if immunizations are not up-to-date. The need for rabies postexposure prophylaxis should also be assessed.

Which interventions decrease the risk for cellulitis and soft tissue infections associated with surgical wounds?

Specific antibiotic prophylaxis should be administered as indicated by the type of surgical procedure within 1 hour before incision; for long procedures, a second dose should be given

after 2 drug half-lives. Postoperative prophylactic antibiotics should not be continued beyond 24 hours. Patients should be advised to shower or bathe with soap or an antiseptic agent at least on the night before surgery. Strict adherence to aseptic technique, including surgical scrubbing and double-gloving, is essential. Skin should be cleaned thoroughly with an alcohol-containing preparatory agent (combined with povidone-iodine or chlorhexidine). Hair should not be removed unless it could interfere with the procedure; if necessary, it should be removed outside of the operating room by clipping. Normothermia should be maintained throughout the perioperative period, and tissue oxygenation should be optimized through supplemental oxygen. Attention to surgical technique, including careful handling of tissue and eradicating dead space, is also recommended (12). For *Staphylococcus aureus*-colonized patients having high-risk procedures (including orthopedic surgery with implantation of hardware and cardiac surgery), guidelines recommend targeted decolonization before surgery (12, 13).

When should *Staphylococcus aureus* decolonization be considered to prevent SSTIs?

Colonization with *S aureus* has been shown to increase risk for subsequent infection, and decolonization protocols can successfully eradicate these bacteria (14). However, preemptive decolonization with mupirocin has not been shown to decrease risk for subsequent infection in either MRSA-colonized healthy (15) or hospitalized (16) persons, and there are concerns regarding emergence of resistance and selection for more virulent strains. Therefore, routine decolonization of methicillin-sensitive *S aureus*

(MSSA) or MRSA is not recommended. However, intensive decolonization may reduce the rate of recurrent SSTIs. Guidelines from the IDSA recommend considering a 5-day decolonization regimen for recurrent *S aureus* SSTIs, consisting of intranasal

mupirocin twice a day, and daily chlorhexidine washes in addition to daily decontamination of personal items (7). Household decolonization should also be considered in the setting of ongoing transmission among household members (17).

Prevention... Prevention of cellulitis and soft tissue infections relies on management of predisposing factors (including vascular insufficiency, lymphedema, and cracked or macerated skin); thorough wound cleansing; and, when appropriate, antimicrobial prophylaxis. For patients with diabetes, attention to foot care is critical. Prevention of surgical site infections focuses on maintenance of antisepsis, prophylactic perioperative antibiotics, and good surgical technique. *S aureus* decolonization should be considered for patients with recurrent purulent SSTIs.

CLINICAL BOTTOM LINE

What is the role of the history and physical examination in the diagnosis of cellulitis and soft tissue infections?

Cellulitis is a skin infection involving deep dermis and subcutaneous fat and is often diagnosed on the basis of a careful history and targeted physical examination. Misdiagnosis is common and leads to antibiotic overuse, unnecessary hospitalizations, drug-related adverse events, and cost (18) (**Table 2**). The history should include questions regarding predisposing conditions and risk factors that may increase the likelihood of specific pathogens (see the **Box: Clinical Diagnoses and Associated Pathogens**). The presentation is typically acute and is almost always unilateral.

It is important to differentiate among SSTIs (**Table 1** and **Figure**). Nonpurulent SSTIs include cellulitis without a purulent focus (which classically presents with localized pain, erythema, induration, and warmth) and erysipelas (where infection is limited to the dermis and epidermis and pres-

ents as well-demarcated and plaque-like with a peau d'orange appearance). These must be distinguished from purulent SSTIs, which include folliculitis, furuncles, carbuncles, and abscesses, for selection of an appropriate antimicrobial agent. Pyomyositis is a deeper purulent infection involving skeletal muscle and is most frequently characterized by systemic signs of infection and cramping pain localized to a muscle group. It may present without superficial signs of involvement. Nonpurulent SSTIs are most likely to be caused by streptococci, whereas purulent SSTIs are usually caused by *S aureus*. Impetigo most commonly begins with bullae or vesicles that progress to crusted lesions, often on the face, and can be caused by either staphylococci or streptococci. Ecthyma is a variant of impetigo that involves deeper levels of the epidermis and progresses to "punched-out"-appearing ulcers. It is most frequently caused by group A streptococci.

Diagnosis

20. Wong CH, Khin LW, Heng KS, Tan KC, Low CO. The LRINEC (Laboratory Risk Indicator for Necrotizing Fasciitis) score: a tool for distinguishing necrotizing fasciitis from other soft tissue infections. *Crit Care Med*. 2004;32:1535-41. [PMID: 15241098]
21. Burner E, Henderson SO, Burke G, Nakashioya J, Hoffman JR. Inadequate Sensitivity of Laboratory Risk Indicator to Rule Out Necrotizing Fasciitis in the Emergency Department. *West J Emerg Med*. 2016;17:333-6. [PMID: 27330667]
22. Murphy G, Markeson D, Choa R, Armstrong A. Raised serum lactate: a marker of necrotizing fasciitis? *J Plast Reconstr Aesthet Surg*. 2013;66:1712-6. [PMID: 23911720]
23. Torres J, Avalos N, Echols L, Mongelluzzo J, Rodriguez RM. Low yield of blood and wound cultures in patients with skin and soft tissue infections. *Am J Emerg Med*. 2017;35:1159-1161. [PMID: 28592371]
24. Crisp JG, Takhar SS, Moran GJ, Krishnadasan A, Dowd SE, Finegold SM, et al; EMERGENCY ID Net Study Group. Inability of polymerase chain reaction, pyrosequencing, and culture of infected and uninfected site skin biopsy specimens to identify the cause of cellulitis. *Clin Infect Dis*. 2015;61:1679-87. [PMID: 26240200]
25. Alsaawi A, Alrajhi K, Alshehri A, Ababtain A, Alsolamy S. Ultrasonography for the diagnosis of patients with clinically suspected skin and soft tissue infections: a systematic review of the literature. *Eur J Emerg Med*. 2017;24:162-169. [PMID: 26485694]
26. Martinez M, Peponis T, Hage A, Yeh DD, Kaafarani HMA, Fagenholz PJ, et al. The Role of Computed Tomography in the Diagnosis of Necrotizing Soft Tissue Infections. *World J Surg*. 2017. [PMID: 28762168]
27. Koning S, van der Wouden JC, Chosidow O, Twynholm M, Singh KP, Scangarella N, et al. Efficacy and safety of retapamulin ointment as treatment of impetigo: randomized double-blind multicentre placebo-controlled trial. *Br J Dermatol*. 2008;158:1077-82. [PMID: 18341664]

Clinical Diagnoses and Associated Pathogens

Diabetes: *Staphylococcus aureus*, group B streptococci, anaerobes, gram-negative bacilli

Cirrhosis: *Campylobacter fetus*, coliforms, *Vibrio vulnificus*, *Capnocytophaga canimorsus*

Neutropenia: *Pseudomonas aeruginosa*

Human bite: *Eikenella corrodens*, viridans group streptococci

Cat bite: *Pasteurella multocida*

Dog bite: *Pasteurella multocida*, *Capnocytophaga canimorsus*

Rat bite: *Streptobacillus moniliformis*

Hot tub exposure: *Pseudomonas aeruginosa*, atypical mycobacteria

Laceration occurring in fresh water: *Aeromonas hydrophila*

Laceration occurring in brackish water: *Vibrio* species

Fish tank exposure: *Mycobacterium marinum*

Fish handling: *Erysipelothrix rhusiopathiae*

Penetrating foot wounds: *Pseudomonas aeruginosa*

Intravenous drug use: Methicillin-resistant *Staphylococcus aureus*, group A streptococci

Necrotizing fasciitis

- Type 1 (polymicrobial, mixed anaerobes and aerobes): Streptococci, *Clostridium* species, *Bacteroides* species, Enterobacteriaceae, staphylococci, enterococci
- Type 2 (monomicrobial): Group A streptococci (most common), community-associated methicillin-resistant *Staphylococcus aureus*, *Clostridium* species (infrequent), *Vibrio* species

Table 2. Differential Diagnosis of Cellulitis and Soft Tissue Infections

Disease	Clinical Features
Infectious	
Erythema migrans	Well-demarcated round plaque, painless, progresses more slowly than cellulitis/erysipelas
Septic arthritis/bursitis	Erythema overlying joint or bursa, exquisitely tender, fever; for septic arthritis, severe pain on range of motion
Early herpes zoster	Dermatomal; erythema and severe pain precede development of vesicles
Streptococcal/staphylococcal toxic shock	Fever, shock, and multiorgan dysfunction, which may be accompanied by diffuse erythroderma mimicking bacterial SSTI
Noninfectious	
Venous stasis dermatitis	Often bilateral, superficial desquamation, weeping/crusting, pitting edema, associated with other features of chronic venous insufficiency
Deep venous thrombosis	Deep pain and swelling, most often in the calf; less frequently associated with marked erythema
Contact dermatitis	Pruritic, history of exposure to irritant, well-defined borders matching area of exposure
Lipodermatosclerosis	Fibrosing panniculitis, painful, poorly demarcated, begins medial to the ankle, red-purple in acute phase (develops over weeks)
Lymphedema	Nonpitting edema, erythema; absence of warmth; secondary skin changes: hyperkeratosis, nodules, hyperpigmentation
Pyoderma gangrenosum	Painful ulcers with well-defined borders, undermining of ulcer edge; often accompanied by systemic illness

SSTI = skin and soft tissue infection.

Signs indicating the possibility of necrotizing SSTIs, such as gas gangrene or necrotizing fasciitis, should be sought as these infections are life- and limb-threatening, are rapidly progressive, and require urgent surgical intervention. Early diagnosis is challenging, and clinicians should have a high index of suspicion for necrotizing SSTIs, particularly in patients who present with systemic signs of systemic toxicity or who have a history of diabetes mellitus, obesity, immunosuppression, hematologic cancer, injection drug use, or the presence of a foreign body (19). Typical signs of necrotizing fasciitis include pain disproportionate to physical examination findings, blisters and bullae, skin discoloration, anesthesia of the skin, or crepitus. However, these may be late findings and lack sensitivity; therefore, their absence should

not dissuade clinicians from surgical exploration in the appropriate clinical setting (see the **Box: Findings That Suggest Necrotizing Fasciitis**).

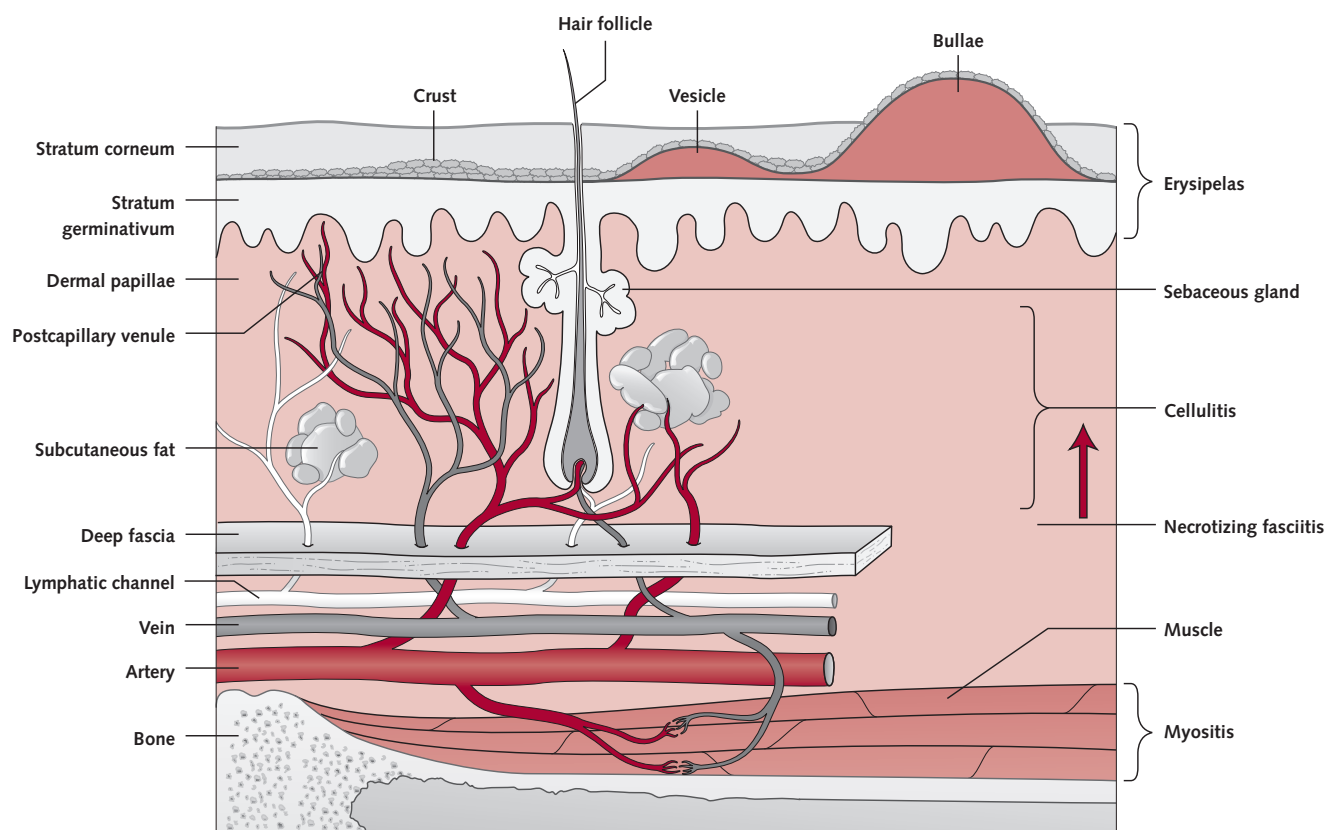
Pyomyositis is most frequently characterized by systemic signs of infection and cramping pain localized to a muscle group and may present without superficial signs of involvement.

What is the role of laboratory testing in diagnosis?

Laboratory testing is not required for the evaluation of cellulitis and uncomplicated soft tissue infections in the absence of comorbidities. Leukocytosis and elevated inflammatory markers may be present but are nonspecific (**Table 3**). Laboratory evaluation is useful to evaluate the severity of infection. The Laboratory Risk Indicator for Necrotizing Fasciitis

28. Bruun T, Oppegaard O, Kittang BR, Mylvaganam H, Langeland N, Skrede S. Etiology of Cellulitis and Clinical Prediction of Streptococcal Disease: A Prospective Study. Open Forum Infect Dis. 2016; 3:ofv181. [PMID: 26734653]

Figure. Anatomy and types of skin and soft tissue infections.



Score, which uses readily available laboratory data from the complete blood count and basic metabolic panel plus C-reactive protein, was developed to risk-stratify patients with cellulitis for necrotizing fasciitis (20). However, subsequent studies have found that this score has inadequate sensitivity to rule out necrotizing fasciitis; thus, a low score should not trump clinical judgment (21). Elevated serum lactate levels have been shown to be strongly associated with tissue necrosis and may be a useful adjunct in the diagnosis of necrotizing infection (22).

Bacteremia is uncommon in the setting of uncomplicated cellulitis and soft tissue infections (23). Blood cultures are therefore not routinely recommended except for patients with immunodeficiency

(those with cancer receiving chemotherapy, neutropenia, severe cell-mediated immunodeficiency, splenectomy), special exposures (animal bites, water-associated injuries), underlying lymphedema, suspected necrotizing infection, or sepsis. Superficial wound swabs and surface cultures represent colonizing microbes and are unlikely to identify the underlying cause. For nonpurulent cellulitis and erysipelas, needle aspiration and punch biopsy cultures are low-yield. Whether positive culture results provide a definitive diagnosis of the cause is unclear because similar rates of *S aureus*-positive cultures have been obtained from infected and noninfected skin. Advanced molecular techniques, including polymerase chain reaction and

29. Moran GJ, Krishnadasan A, Mower WR, Abrahamian FM, LoVecchio F, Steele MT, et al. Effect of Cephalexin Plus Trimethoprim-Sulfamethoxazole vs Cephalexin Alone on Clinical Cure of Uncomplicated Cellulitis: A Randomized Clinical Trial. *JAMA*. 2017;317:2088-2096. [PMID: 28535235]
30. Miller LG, Daum RS, Creech CB, Young D, Downing MD, Eells SJ, et al; DMD 07-0051 Team. Clindamycin versus trimethoprim-sulfamethoxazole for uncomplicated skin infections. *N Engl J Med*. 2015;372:1093-103. [PMID: 25785967]

Findings That Suggest Necrotizing Fasciitis

- Pain out of proportion to physical examination findings
- Edema, induration, or pain beyond area of apparent skin involvement
- Violaceous blisters or bullae
- Pale or mottled skin
- Anesthesia of the skin
- Crepitus
- Skin necrosis or ecchymosis
- Rapid progression
- Failure to respond to initial antibiotics
- Systemic toxicity
- Multiorgan failure
- Incision into necrotic soft tissues yields no bleeding

31. Talan DA, Lovecchio F, Abrahamian FM, Karras DJ, Steele MT, Rothman RE, et al. A Randomized Trial of Clindamycin Versus Trimethoprim-sulfamethoxazole for Uncomplicated Wound Infection. *Clin Infect Dis*. 2016;62:1505-1513. [PMID: 27025829]

32. Shorr AF, Lodise TP, Corey GR, De Anda C, Fang E, Das AF, et al. Analysis of the phase 3 ESTABLISH trials of tedizolid versus linezolid in acute bacterial skin and skin structure infections. *Antimicrob Agents Chemother*. 2015;59:864-71. [PMID: 25421472]

33. McCurdy S, Lawrence L, Quintas M, Woosley L, Flamm R, Tseng C, et al. In Vitro Activity of Delafloxacin and Microbiological Response against Fluoroquinolone-Susceptible and Nonsusceptible *Staphylococcus aureus* Isolates from Two Phase 3 Studies of Acute Bacterial Skin and Skin Structure Infections. *Antimicrob Agents Chemother*. 2017;61. [PMID: 28630189]

34. Corey GR, Kabler H, Mehra P, Gupta S, Overcash JS, Porwal A, et al; SOLO I Investigators. Single-dose oritavancin in the treatment of acute bacterial skin infections. *N Engl J Med*. 2014; 370:2180-90. [PMID: 24897083]

35. Boucher HW, Wilcox M, Talbot GH, Puttagunta S, Das AF, Dunne MW. Once-weekly dalbavancin versus daily conventional therapy for skin infection. *N Engl J Med*. 2014; 370:2169-79. [PMID: 24897082]

Table 3. Laboratory and Other Studies for Evaluating Cellulitis and Soft Tissue Infections

Complete blood count and differential	Elevated leukocyte count nonspecific but suggestive of infection; markedly elevated counts with left shift suggestive of deep-seated or aggressive infection. Thrombocytopenia suggestive of more severe infection. Hemoconcentration and leukemoid reaction have been described in <i>Clostridium sordellii</i> infection, and <i>Clostridium perfringens</i> infection has been associated with hemolysis
Lactate	Elevated levels indicate presence of tissue/organ underperfusion (e.g., sepsis) and/or possible tissue necrosis
Creatine phosphokinase	Elevated levels indicate myonecrosis and may indicate necrotizing fasciitis. Also frequently elevated in <i>Vibrio vulnificus</i> infection
C-reactive protein	Nonspecific, may help risk-stratify patients with suspected NSTI
Gram stain and culture	Should be obtained to guide therapy in purulent infections and if surgical debridement is performed
Radiography	May show gas in tissues but poorly sensitive for identifying NSTI
CT with contrast	May be useful for identifying NSTI when the diagnosis is uncertain
MRI	Test of choice for pyomyositis and osteomyelitis; poor specificity for NSTI
Ultrasonography	Useful for identifying drainable fluid collections/abscesses

CT = computed tomography; MRI = magnetic resonance imaging; NSTI = necrotizing soft tissue infection.

pyrosequencing of skin biopsy specimens, have been similarly disappointing (24). Routine culture of simple uncomplicated abscesses is probably not necessary but should be done in cases of recurrent infection, treatment failure, atypical features, and severe disease. If infected tissues are debrided, deep intraoperative cultures should be obtained. Infected bite wounds should also be cultured.

What is the role of imaging in diagnosis?

Ultrasonography can be useful for distinguishing nonpurulent cellulitis from cellulitis with underlying abscess and for identifying drainable fluid collection, particularly if the clinical assessment is indeterminate (25). Magnetic resonance imaging is the test of choice for identifying pyomyositis or underlying osteomyelitis. For diagnosis of necrotizing

fasciitis, direct surgical examination of the fascia is the gold standard and surgery should not be delayed to obtain imaging studies. Plain films may show gas in the soft tissues in the presence of necrotizing soft tissue infection (NSTI) but are insensitive. Magnetic resonance imaging has not been well-studied for identification of necrotizing fasciitis, and the time required to obtain these studies limits its utility in this setting. Intravenous contrast-enhanced computed tomography (CT) has shown promise for detection of NSTI.

One recent, retrospective, single-center study of 184 patients who had contrast-enhanced CT scans before surgical exploration demonstrated a sensitivity of 100% and specificity of 98% for diagnosis of NSTI. However, an important limitation is that patients with a high suspicion of NSTI would have been taken directly to surgery and excluded from the study (26).

Thus, CT with contrast may be useful for less critically ill patients in whom the diagnosis is uncertain but should not be relied on for exclusion.

When should clinicians consult an interventional radiologist or surgeon?

Patients with any clinical suspicion for NSTI, including those presenting with cellulitis and

shock, should be promptly evaluated by a surgeon because any delay in recognition or definitive surgical management increases the risk for mortality. Interventional radiologists can use imaging guidance to perform biopsies or aspirate deep collections for pathology and culture and to perform therapeutic drainage.

Diagnosis... Cellulitis is a clinical diagnosis made on the basis of history and physical examination. Many conditions mimic cellulitis; thus, rates of misdiagnosis are high. Laboratory testing is not required for evaluation of uncomplicated infections but is useful to determine disease severity. Clinicians should be watchful for signs and symptoms of necrotizing infection, such as severe pain, rapid progression, and systemic toxicity. Imaging should not delay surgical intervention for necrotizing fasciitis. Ultrasonography may be useful to identify abscesses.

CLINICAL BOTTOM LINE

Which adjuvant measures are helpful in the treatment of patients with cellulitis and soft tissue infection?

For cellulitis, elevation of the affected limb to facilitate drainage may hasten improvement. Treatment of underlying cutaneous disorders, such as fissure, macerated skin, or tinea pedis, is recommended. Hyperbaric oxygen has been investigated as an adjunctive treatment of diabetic foot infections and gas gangrene; however, it is not recommended because benefit has not been proven and it may delay critical treatments, including surgical debridement.

When is topical antimicrobial therapy appropriate?

Either topical antimicrobial therapy, with mupirocin or retapamulin (twice daily for 5 days) (27), or oral therapy is recommended to treat localized bullous or non-bullous impetigo, but oral ther-

apy is recommended for patients with numerous lesions or ecthyma. Oral therapy is also recommended during outbreaks of impetigo to reduce person-to-person transmission or with post-streptococcal glomerulonephritis to reduce circulation of nephritogenic strains.

How should clinicians determine whether to prescribe oral or parenteral antimicrobials?

Oral therapy is preferred for uncomplicated cellulitis and other SSTIs without systemic toxicity. Patients with moderate to severe infection, including those with fever or other signs of systemic illness, evidence of end-organ dysfunction, or clinical signs of deeper infection (including pyomyositis, necrotizing infection, and gangrene) require parenteral antibiotics. Facial erysipelas should be treated initially with parenteral antibiotics, and

Treatment

36. Dunne MW, Puttagunta S, Giordano P, Krievins D, Zelasky M, Baldassarre J. A Randomized Clinical Trial of Single-Dose Versus Weekly Dalbavancin for Treatment of Acute Bacterial Skin and Skin Structure Infection. *Clin Infect Dis*. 2016;62:545-51. [PMID: 26611777]
37. Talan DA, Mower WR, Krishnadasan A. Trimethoprim-Sulfamethoxazole for Uncomplicated Skin Abscess [Letter]. *N Engl J Med*. 2016;375:285-6. [PMID: 27468069]
38. Daum RS, Miller LG, Immergluck L, Fritz S, Creech CB, Young D, et al; DMID 07-0051 Team. A Placebo-Controlled Trial of Antibiotics for Smaller Skin Abscesses. *N Engl J Med*. 2017;376:2545-2555. [PMID: 28657870]
39. Jenkins TC, Knepper BC, Jason Moore S, Savelli CC, Pawlowski SW, Perlman DM, et al. Comparison of the microbiology and antibiotic treatment among diabetic and nondiabetic patients hospitalized for cellulitis or cutaneous abscess. *J Hosp Med*. 2014;9:788-94. [PMID: 25266293]
40. Albrecht VS, Limbago BM, Moran GJ, Krishnadasan A, Gorwitz RJ, McDougal LK, et al; EMERGENCY ID NET Study Group. Staphylococcus aureus Colonization and Strain Type at Various Body Sites among Patients with a Closed Abscess and Uninfected Controls at U.S. Emergency Departments. *J Clin Microbiol*. 2015;53:3478-84. [PMID: 26292314]
41. Talan DA, Krishnadasan A, Gorwitz RJ, Fosheim GE, Limbago B, Albrecht V, et al; EMERGENCY ID Net Study Group. Comparison of Staphylococcus aureus from skin and soft tissue infections in US emergency department patients, 2004 and 2008. *Clin Infect Dis*. 2011;53:144-9. [PMID: 21690621]
42. Moran GJ, Krishnadasan A, Gorwitz RJ, Fosheim GE, McDougal LK, Carey RB, et al; EMERGENCY ID Net Study Group. Methicillin-resistant S. aureus infections among patients in the emergency department. *N Engl J Med*. 2006;355:666-74. [PMID: 16914702]

Risk Factors for MRSA

- History of MRSA infection
- Nasal inhalation or smoking of illegal drugs or injection drug use
- Recent incarceration
- Contact sports
- Frequent visits to bars, raves, or clubs
- HIV infection
- Recent antibiotic use
- Recent hospitalization
- Hemodialysis
- Close contact with known or suspected MRSA infection

MRSA = methicillin-resistant *Staphylococcus aureus*.

antibiotics should also be considered for patients with immunodeficiency and those in whom oral therapy has failed.

In patients for whom oral therapy is appropriate, how should clinicians choose a specific antimicrobial agent?

β -hemolytic streptococci are believed to cause the vast majority of cases of cellulitis (28). Thus, for nonpurulent cellulitis, an agent covering these bacteria should be chosen. Acceptable agents include penicillin, amoxicillin, dicloxacillin, cephalexin, or clindamycin for patients with severe penicillin allergy (**Table 4**). For otherwise-healthy outpatients with uncomplicated cellulitis, MRSA coverage is not necessary. However, it should be considered for patients with a history of penetrating trauma, injection drug use, known MRSA colonization, evidence of MRSA infection elsewhere, or a close contact with a history of MRSA infection (see the **Box**: Risk Factors for MRSA).

A multicenter, randomized, double-blind trial of cephalexin versus cephalexin plus trimethoprim-sulfamethoxazole (TMP-SMX) for outpatient treatment of cellulitis without associated abscess or purulent discharge showed that adding TMP-SMX did not improve clinical cure rates, which were 83.5% with TMP-SMX and 85.5% without ($P = 0.5$). This trial excluded patients with underlying skin conditions, history of intravenous drug use with fever, and immunosuppression; 10.9% of the participants had diabetes (29).

Although TMP-SMX has historically been considered inadequate as monotherapy for nonpurulent cellulitis due to apparent reduced susceptibility of *Streptococcus pyogenes* to sulfa antibiotics in vitro, more recent clinical data suggest that it is an effective single agent for treating uncomplicated SSTIs, including the subgroup of patients with nonpurulent cellulitis.

In a multicenter trial of 524 outpatients with uncomplicated SSTIs (cellulitis, abscess, or both) that compared clindamycin with TMP-SMX, cure rates were similar (80.3% for clindamycin vs. 77.7% for TMP-SMX [$P = 0.52$] in the intention-to-treat analysis and 89.5% vs. 88.2% [$P = 0.77$] in the per protocol analysis)

Table 4. Antimicrobials Commonly Used for Oral Therapy for Cellulitis and Soft Tissue Infections

Agent	Usual Dosage*	Comments
Streptococci		
Amoxicillin	500 mg PO TID	Useful for <i>Pasteurella</i> species, better bioavailability than penicillin
Penicillin VK	500 mg PO QID	Narrow spectrum; frequent dosing
Streptococci and MSSA		
Amoxicillin-clavulanate	875/125 mg PO BID	Includes anaerobic coverage
Dicloxacillin	500 mg PO QID	Frequent dosing
Cephalexin	500 mg PO QID	Frequent dosing
MRSA (and streptococci/MSSA)		
Clindamycin	300 mg PO BID	Greatest association with secondary <i>Clostridium difficile</i> infection
Doxycycline	100 mg PO BID	Causes photosensitivity; fewer clinical data
Trimethoprim-sulfamethoxazole	1 double-strength tablet PO BID	Can cause hyperkalemia; use caution in patients with impaired renal function or on angiotensin-converting enzyme inhibitors/angiotensin-receptor blockers
Linezolid	600 mg PO BID	Risk for serotonin syndrome with concomitant selective serotonin reuptake inhibitors; bone marrow toxicity with prolonged use
Tedizolid	200 mg PO BID	Expensive; less risk for thrombocytopenia and drug interactions than linezolid
Delafloxacin	450 mg PO BID	Limited clinical experience

BID = twice daily; MRSA = methicillin-resistant *Staphylococcus aureus*; MSSA = methicillin-sensitive *Staphylococcus aureus*; PO = oral; QID = 4 times daily; TID = 3 times daily.

* May vary on the basis of weight, renal function, and indication.

(30). Similar results were seen in a trial of clindamycin versus TMP-SMX for treatment of wound infection (31).

For impetigo, in the absence of culture data treatment should target both streptococci and staphylococci. For purulent infections, including abscesses, furuncles, and carbuncles with or without associated cellulitis (which are most often caused by staphylococci), an agent that is active against MRSA is recommended. Oral antibiotics for MRSA include TMP-SMX, doxycycline, clindamycin, and linezolid. Susceptibility of MRSA isolates to clindamycin varies geographically; thus, it should not be used as first-line empirical MRSA coverage if local resistance rates are high. Novel oral antibiotics have recently been approved for the treatment of SSTIs. Tedizolid is a novel oxazolidinone, which was approved in June 2014. Like linezolid, it offers broad activity against gram-positive pathogens, is available both orally and parenterally, and seems to have less hematologic toxicity and fewer drug interactions (32). Delafloxacin, a novel broad-spectrum fluoroquinolone, was recently approved by the U.S. Food and Drug Administration for treatment of acute bacterial SSTIs and seems to be effective against even levofloxacin-nonsusceptible MRSA isolates (33).

In patients for whom parenteral therapy is appropriate, how should clinicians choose a specific antimicrobial regimen?

As with oral therapy, treatment of moderate nonpurulent cellulitis should focus on β -hemolytic streptococci. Vancomycin, or another antimicrobial agent covering MRSA (**Table 5**) and streptococci, should be used for purulent infections requiring parenteral therapy. These agents should also be used for

nonpurulent cellulitis in patients at high risk for MRSA infection, such as those with a history of intravenous drug use, proximity to an indwelling catheter, penetrating trauma, or MRSA colonization. Adding empirical broad-spectrum, gram-negative coverage should be considered for patients with severe and rapidly progressing infections, perirectal infection, or severe immunocompromise.

The novel semisynthetic lipoglycopeptides dalbavancin and oritavancin are active against a wide spectrum of gram-positive pathogens, including MRSA, and have very long terminal serum half-lives. For SSTIs, dalbavancin can be given as a single dose over 30 minutes (34), and oritavancin can be given as a single 3-hour infusion (35, 36). Thus, they can be considered for patients who require parenteral therapy but decline hospitalization or for those who may be unlikely to adhere to an oral regimen.

What is appropriate antimicrobial therapy for infections associated with human or animal bites?

Infected human bite wounds are often polymicrobial with both aerobic bacteria (including viridans streptococci, *Eikenella corrodens*, and staphylococci) and anaerobes (including *Bacteroides* species, *Fusobacterium* species, and peptostreptococci). Infected animal bite wounds often involve *Pasteurella* species in addition to anaerobes, streptococci, and *S aureus*. *Eikenella* and *Pasteurella* species are both resistant to clindamycin and have decreased susceptibility to first-generation cephalosporins. The treatment of choice for both infected human and animal bite wounds is amoxicillin-clavulanate for oral therapy or ampicillin-sulbactam for intravenous therapy (with addition of vancomycin for severe

43. Ray GT, Suaya JA, Baxter R. Incidence, microbiology, and patient characteristics of skin and soft tissue infections in a U.S. population: a retrospective population-based study. *BMC Infect Dis*. 2013;13:252. [PMID: 23721377]
44. Liu C, Bayer A, Cosgrove SE, Daum RS, Fridkin SK, Gorwitz RJ, et al; Infectious Diseases Society of America. Clinical practice guidelines by the infectious diseases society of america for the treatment of methicillin-resistant *Staphylococcus aureus* infections in adults and children. *Clin Infect Dis*. 2011;52:e18-55. [PMID: 21208910]
45. Miller LG, Perdreau-Remington F, Bayer AS, Diep B, Tan N, Bharadwaj K, et al. Clinical and epidemiologic characteristics cannot distinguish community-associated methicillin-resistant *Staphylococcus aureus* infection from methicillin-susceptible *S. aureus* infection: a prospective investigation. *Clin Infect Dis*. 2007;44: 471-82. [PMID: 17243048]
46. Dantes R, Mu Y, Belflower R, Aragon D, Dumyati G, Harrison LH, et al; Emerging Infections Program-Active Bacterial Core Surveillance MRSA Surveillance Investigators. National burden of invasive methicillin-resistant *Staphylococcus aureus* infections, United States, 2011. *JAMA Intern Med*. 2013;173:1970-8. [PMID: 24043270]
47. Kadri SS, Swihart BJ, Bonne SL, Hohmann SF, Hennessy LV, Louras P, et al. Impact of Intravenous Immunoglobulin on Survival in Necrotizing Fasciitis With Vasopressor-Dependent Shock: A Propensity Score-Matched Analysis From 130 US Hospitals. *Clin Infect Dis*. 2017;64: 877-885. [PMID: 28034881]

Table 5. Antimicrobials Commonly Used for Parenteral Therapy of Cellulitis and Soft Tissue Infections

Agent	Usual Dosage	Comments
Streptococci		
Penicillin G	2–4 million units IV every 4–6 h	First line for group A streptococci
Ceftriaxone	1–2 g IV every 24 h	Has good gram-negative coverage (does not cover <i>Pseudomonas</i> species or extended-spectrum β -lactamase-producing Enterobacteriaceae)
Streptococci and MSSA		
Cefazolin	1 g IV every 8 h	Has some gram-negative coverage
Nafcillin	1–2 g IV every 4 h	More adverse reactions than cefazolin (rash, drug fever, cytopenias)
MRSA (and streptococci/MSSA)		
Vancomycin	15 mg/kg IV every 12 h	Requires monitoring of troughs; can cause "red man syndrome" and nephrotoxicity
Daptomycin	4 mg/kg IV every 24 h	Higher dose required for bacteremia, requires monitoring of creatine phosphokinase levels
Linezolid	600 mg IV every 12 h	Risk for serotonin syndrome with concomitant selective serotonin reuptake inhibitors; bone marrow toxicity with prolonged use
Clindamycin	600–900 mg IV every 8 h	Greatest association with secondary <i>Clostridium difficile</i> infection; increasing rates of resistant <i>Staphylococcus aureus</i>
Ceftaroline	600 mg IV every 12 h	Has gram-negative coverage similar to ceftriaxone
Dalbavancin	One dose of 1500 mg IV over 30 min	Single-dose regimen found to be noninferior to 2-dose regimen; limited clinical experience
Oritavancin	One dose of 1200 mg IV over 3 h	Artificially prolongs prothrombin time and partial thromboplastin time; limited clinical experience

IV = intravenous; MRSA = methicillin-resistant *Staphylococcus aureus*; MSSA = methicillin-sensitive *Staphylococcus aureus*.

infections until MRSA involvement is ruled out). For penicillin-allergic patients, levofloxacin plus clindamycin or metronidazole or moxifloxacin is appropriate for both infected animal and human bite wounds. Deep or severe infections, or those involving the hand, should initially be treated with intravenous antibiotics.

Should antibiotics be given as an adjunct to incision and drainage for small skin abscesses?

The IDSA guidelines, published in 2014 (7), recommend adjunctive antibiotics (active against MRSA) only for patients with signs of systemic illness or those who are immunocompromised and consideration of antibiotics for patients with multiple abscesses, extremes of age, or lack of response to incision and drainage alone. However, 2 recent clinical trials have demonstrated

that adjunctive antibiotics increase clinical cure rates for abscesses, including smaller ones (<5 cm) (37, 38).

A large, multicenter, randomized trial including 1247 participants compared TMP-SMX with placebo (7 days) in patients with cutaneous abscesses measuring at least 2 cm being treated with drainage. Cure rates were higher in the TMP-SMX group (92.9% vs. 85.7% in the per protocol analysis [$P < 0.001$]). Percentage of abscesses requiring subsequent drainage, infection rates at new sites, and infection rates in household members were all lower in the treatment group (37).

In a subsequent trial, 786 patients with skin abscesses smaller than 5 cm in diameter were randomly assigned to receive TMP-SMX, clindamycin, or placebo for 10 days after incision and drainage. The cure rate for patients treated with TMP-SMX or clindamycin was significantly higher than for those treated with placebo (83.1% for TMP-SMX and 81.7% for clindamycin vs. 68.9% for placebo [$P < 0.001$ for both comparisons]). Rates of adverse events were similar in the TMP-SMX (11.1%) and placebo (12.5%) groups and higher in the clindamycin group (21.9%). One hypersensitivity reaction was noted in the TMP-SMX group (38).

48. Madsen MB, Hjortrup PB, Hansen MB, Lange T, Norrby-Teglund A, Hyldegaard O, et al. Immunoglobulin G for patients with necrotising soft tissue infection (IN-STINCT): a randomised, blinded, placebo-controlled trial. *Intensive Care Med*. 2017;43:1585-1593. [PMID: 28421246]

49. Hepburn MJ, Dooley DP, Skidmore PJ, Ellis MW, Starnes WF, Hasewinkle WC. Comparison of short-course (5 days) and standard (10 days) treatment for uncomplicated cellulitis. *Arch Intern Med*. 2004;164:1669-74. [PMID: 15302637]

Thus, adjunctive antibiotics should be considered for all patients with uncomplicated skin abscesses, although incision and drainage may be sufficient for small abscesses, and the risks and benefits should be weighed for each individual.

Are there special considerations for immunocompromised hosts?

Immunocompromised patients are predisposed to infection with opportunistic pathogens including fungi, viruses, and mycobacteria. Furthermore, patients with underlying immunodeficiency may present with atypical or subtle manifestations and without systemic signs despite severe infection. The approach to diagnosis and treatment of immunocompromised patients should be aggressive. Imaging studies guided by clinical signs and symptoms may be helpful to delineate the extent of infection, and the threshold for skin biopsy—particularly in cases of atypical features or a slow response to therapy—should be low. For neutropenic patients, empirical treatment of both gram-positive organisms, including MRSA, and gram-negative organisms, including *Pseudomonas* species, is recommended and should be instituted at the first signs or symptoms of infection. Ecthyma gangrenosum is a cutaneous infection most frequently caused by *P aeruginosa*. This disorder is classically associated with bacteremia in immunocompromised patients and is characterized by necrotic ulcers (which may also be caused by other gram-negative organisms, as well as MRSA, atypical mycobacteria, and *Fusarium* species). In patients with prolonged neutropenia (particularly if fever is ongoing or recurrent despite antimicrobial therapy), invasive fungal infections, including *Candida*,

Aspergillus, *Mucor*, and *Fusarium* species, are a major cause of morbidity and mortality. Although they less frequently cause primary skin infections, they may present with cutaneous dissemination and empirical treatment should be considered.

Are there special considerations for antimicrobial selection in patients with diabetes mellitus?

For diabetic patients with uncomplicated cellulitis and cutaneous abscesses that are not associated with an infected ulcer, the microbiology seems to be similar to that of nondiabetic patients, with a predominance of gram-positive aerobic bacteria (39). Therefore, for these infections, empirical treatment should be the same as for nondiabetic patients as described above. For mild diabetic foot infections (involving only skin and subcutaneous tissue [< 2 cm around ulcer] without systemic signs of infection) an agent covering *Streptococcus* species and MSSA (or if risk factors are present, MRSA) is recommended. Broad-spectrum coverage, targeting MRSA, anaerobes, and gram-negative rods, is warranted for moderate and severe diabetic foot infections.

When should clinicians consider coverage of MRSA?

The proportion of MRSA isolates from SSTIs in the United States is high (at least 50%) (40–43), and MRSA is the most common identifiable pathogen of SSTIs in patients presenting to U.S. emergency departments (42). Thus, for all patients with purulent SSTIs, empirical coverage for MRSA is recommended pending culture results (44). Numerous risk factors have been identified for community-acquired MRSA (see the **Box**: Risk Factors for MRSA). However, the sensitivity,

Indications for Hospitalization of Patients With Cellulitis and Soft Tissue Infections

- Systemic toxicity
- Evidence of end-organ damage
- Concern for necrotizing infection
- Limb-threatening infection
- Need for surgical intervention (aside from simple bedside irrigation and drainage)
- Concern for inability to adhere to therapy

specificity, and predictive values of these risk factors are poor (45). Although rates of MRSA infection seem to be declining (46), they still represent a large proportion of SSTIs.

When should patients be hospitalized?

Most patients with cellulitis have mild uncomplicated infection and can be treated as outpatients, but patients with moderate to severe SSTIs, including those for whom there is concern for a deeper or necrotizing infection, require inpatient admission and intravenous antibiotics (see the **Box**: Indications for Hospitalization of Patients with Cellulitis and Soft Tissue Infections). Hospitalization should also be considered for patients who are severely immunocompromised, patients for whom there are significant concerns regarding adherence to therapy, and if outpatient treatment is failing. Once the patient shows signs of clinical improvement, is afebrile, and can tolerate oral medications, discharge on an oral antibiotic regimen can be considered.

When should clinicians consider consulting an infectious disease expert?

An infectious disease consultation should be considered for patients with severe cellulitis, including those with necrotizing infection, associated bacteremia, or severe immune compromise, as well as for patients in whom therapy is failing, if unusual pathogens are suspected, or for patients with recurrent cellulitis (for consideration of prophylactic antibiotics).

What should clinicians consider in managing patients with NSTIs?

Type I necrotizing fasciitis is defined as that caused by polymicrobial infection, most commonly mixed aerobes and anaerobes. Type II necrotizing fasciitis is monomicrobial, most commonly due to group A *Streptococcus* but can also be caused by community-associated MRSA or *Vibrio* species. Gas gangrene and myonecrosis are most frequently caused by *Clostridium* species. NSTI is a medical emergency requiring prompt initiation of antibiotics and surgical debridement. Empirical antibiotic therapy should be broad-spectrum, including coverage for MRSA, gram-negative rods, and anaerobes. In addition, because group A streptococcal necrotizing fasciitis cannot reliably be distinguished on clinical grounds, clindamycin should be added to inhibit toxin production. Emergent surgical consultation should be obtained because a delay in the procedure increases the risk for mortality. Tissue cultures should be obtained in the operating room to help guide definitive antimicrobial therapy. Most patients with necrotizing fasciitis require several operations until all necrotic tissue has been debrided. Intravenous immunoglobulin has been investigated as an adjunctive treatment but is not recommended

because no clinical study to date has shown any effect on mortality, hospital length of stay, or physical functioning (47, 48).

What are the indications for surgery?

Early drainage should be performed for all purulent SSTIs, including cutaneous abscesses, carbuncles, large furuncles, and pyomyositis. Incision and drainage for cutaneous abscesses is preferred over needle aspiration. As discussed, necrotizing infections, including gas gangrene, myonecrosis, and necrotizing fasciitis, require urgent surgical debridement. Debridement should also be considered for cutaneous angioinvasive fungal infections. Surgical site infections should be treated with suture removal and incision and drainage.

How long should patients receive antimicrobial therapy?

For cellulitis, the recommended duration of antimicrobial therapy is 5 days (49), with an extension if improvement is not noted. For cutaneous abscesses, 7 days of adjunctive antibiotics is appropriate for most patients. For neutropenic patients with bacterial SSTIs, the recommended treatment duration is 7 to 14 days. The optimal duration for NSTIs has not been defined, but antimicrobial therapy should be continued until no further debridement is required and the patient is clinically stable.

Treatment... Treatment of nonpurulent uncomplicated cellulitis should target streptococci. Empirical coverage for purulent infections should include MRSA. Treatment of abscesses requires drainage, and cure rates are increased with adjunctive antibiotics. NSTIs require emergent surgical evaluation for debridement.

CLINICAL BOTTOM LINE

What factors do U.S. stakeholders use to evaluate the quality of care for soft tissue infection and cellulitis?

Cellulitis and surgical site infections currently have no specific core measures, which are national standards developed by The Joint Commission and the Centers for Medicare & Medicaid Services to assess health care facilities. Surgical site infections are reportable to the National Healthcare Safety Network of the Centers for Disease Control and Prevention.

What do professional organizations recommend regarding the care of patients with cellulitis and soft tissue infection?

The IDSA periodically publishes guidelines on management of SSTIs, the most recent of which were published in 2014. This review largely reflects their recommendations. See the Tool Kit for other guidelines related to the prevention and treatment of soft tissue infections.

In the Clinic Tool Kit

Cellulitis and Soft Tissue Infections

Clinical Guidelines

http://www.idsociety.org/Guidelines/Patient_Care/IDSA_Practice_Guidelines/Infections_by_Organ_System/Skin_Soft_Tissue/Skin_and_Soft_Tissue_Infections/

Clinical practice guidelines from the Infectious Diseases Society of America.

Clinical Resources

<https://www.cdc.gov/mrsa/community/clinicians/index.html>

Detailed information regarding methicillin-resistant *Staphylococcus aureus* from the Centers for Disease Control and Prevention.

https://www.cdc.gov/mrsa/pdf/flowchart_pkterd.pdf

Useful information in single page related to skin and soft tissue infections.

Patient Information

<http://www.aafp.org/afp/2015/0915/p474-s1.html>

Patient information from the American Academy of Family Physicians.

In the Clinic

WHAT YOU SHOULD KNOW ABOUT CELLULITIS AND SOFT TISSUE INFECTIONS

In the Clinic
Annals of Internal Medicine

What Is Cellulitis?

Cellulitis is an infection of the skin and soft tissues below the skin. It occurs when bacteria enter your body through a cut or opening in the skin. You could get cellulitis from:

- A cut or injury that opens your skin, such as through an accident or surgery
- Burns
- Human, animal, or insect bites
- Skin ulcers
- Dermatitis or other skin conditions

Certain health conditions increase your risk for cellulitis, including:

- Diabetes
- Peripheral artery disease
- Cirrhosis
- HIV
- Obesity

What Are the Warning Signs?

- Red, hot, and swollen skin in the affected area
- Pain
- Pus leaking from the wound
- Tight or glossy look to the skin
- Chills and fever
- Skin sores
- Pus or fluid that collects under the skin

How Is It Diagnosed?

The infection may affect only your skin; however, it may also affect tissues under your skin, which could cause infection to spread to your blood and lymph nodes.

- Your doctor will ask you about your symptoms. He or she will also carefully examine your skin to check for signs of infection.
- In rare cases, you may need a blood test.

How Is It Treated?

- It is important to call your doctor right away if you think your skin is infected. If the infection spreads, it can be dangerous.
- Your doctor will clean your skin where it is infected.
- Fluid from the wound may need to be drained.



- You may be given antibiotics to kill the bacteria.
- The infection usually goes away after you take antibiotics. You may need to take them for a longer period if the infection is more severe. If you are given antibiotics, be sure to finish them, even if you feel better or the infection has gone away.

How Can It Be Prevented?

- If you get a minor cut or injury that breaks your skin, be sure to clean it well with soap and water and use an antibiotic ointment, which is available at your local pharmacy.
- If you have a deep wound, a puncture, or any other serious injury, see your doctor. The injury might need to be cleaned using special instruments.
- If you have diabetes, get regular foot examinations and wear diabetic shoes.
- If you are prone to cellulitis, your doctor may prescribe an antibiotic that you will need to take for a longer time.

Questions for My Doctor

- How do I know if I have an infection?
- How did I get an infection?
- How long before the treatment starts to work?
- Are there any side effects from taking antibiotics?
- How can I prevent this from happening again?
- Are there any activities I should avoid?

For More Information



Medline Plus

www.nlm.nih.gov/medlineplus/cellulitis.html

National Institute of Allergy and Infectious Diseases

www.niaid.nih.gov/diseases-conditions/group-strep-types

CORRECTION: INCORRECT DOSAGE IN TABLE

In the February 2018 In the Clinic article (1), the dosage of daptomycin listed in Table 5 (Antimicrobials Commonly Used for Parenteral Therapy of Cellulitis and Soft Tissue Infections) is incorrect. The correct dosage is 4 mg/kg IV every 24 h.

Reference

1. Bystritsky R, Chambers H. Cellulitis and soft tissue infections. *Ann Intern Med.* 2018;168:ITC17-ITC32. [PMID: 29404597] doi:10.7326/AITC201802060